Capstone Design Project I

Progress Report



A Modular Self-Reconfigurable Robot (MSRR) for Search and Space Exploration

Submitted to the
Project Committee appointed by the
International School of Engineering (ISE)
Faculty of Engineering, Chulalongkorn University

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1/2026: 2147498 Capstone Design Project I
Robotics and Artificial Intelligence Engineering (International Program)
International School of Engineering (ISE), Faculty of Engineering,
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1 Project Timeline

- **Completed:** Project concept, objectives, literature review, and phased system architecture established.
- **In progress:** Phase A — single-module locomotion, sensing, communication, and simulation baseline.
- **Next:** Phase B — EP-Face docking, task-oriented reconfiguration, and simple assembled locomotion.

2 Project Phases

2.1 Phase A

External perception and computation. Develop the mechanically homogeneous module, differential-drive locomotion, joint control, odometry, and module-to-commander communication. Validate the module and physics twin before physical docking is introduced.

2.2 Phase B

Docking and task-oriented reconfiguration. Add the electropermanent EP-Face connectors, infrared alignment, and connector-state feedback. Extend the reconfiguration pipeline with task assignment, parallel assembly sequencing, and selection from a library of feasible configurations.

2.3 Phase C

Onboard perception with external computation. Mount a monocular RGB camera and an 8×8 multi-zone ToF sensor on the leader module. Estimate obstacle geometry and traversability locally, while transmitting sensor data to an external computer for processing.

2.4 Phase D

Fully onboard perception and computation. Move the perception and configuration-selection pipeline onto a coordinator module using an embedded compute board. Close the perceive–select–reconfigure–traverse loop on hardware without external localisation or computation.

3 Gantt Chart

Task schedule from the team Notion timeline (as of 25 Sep 2026), grouped by phase.

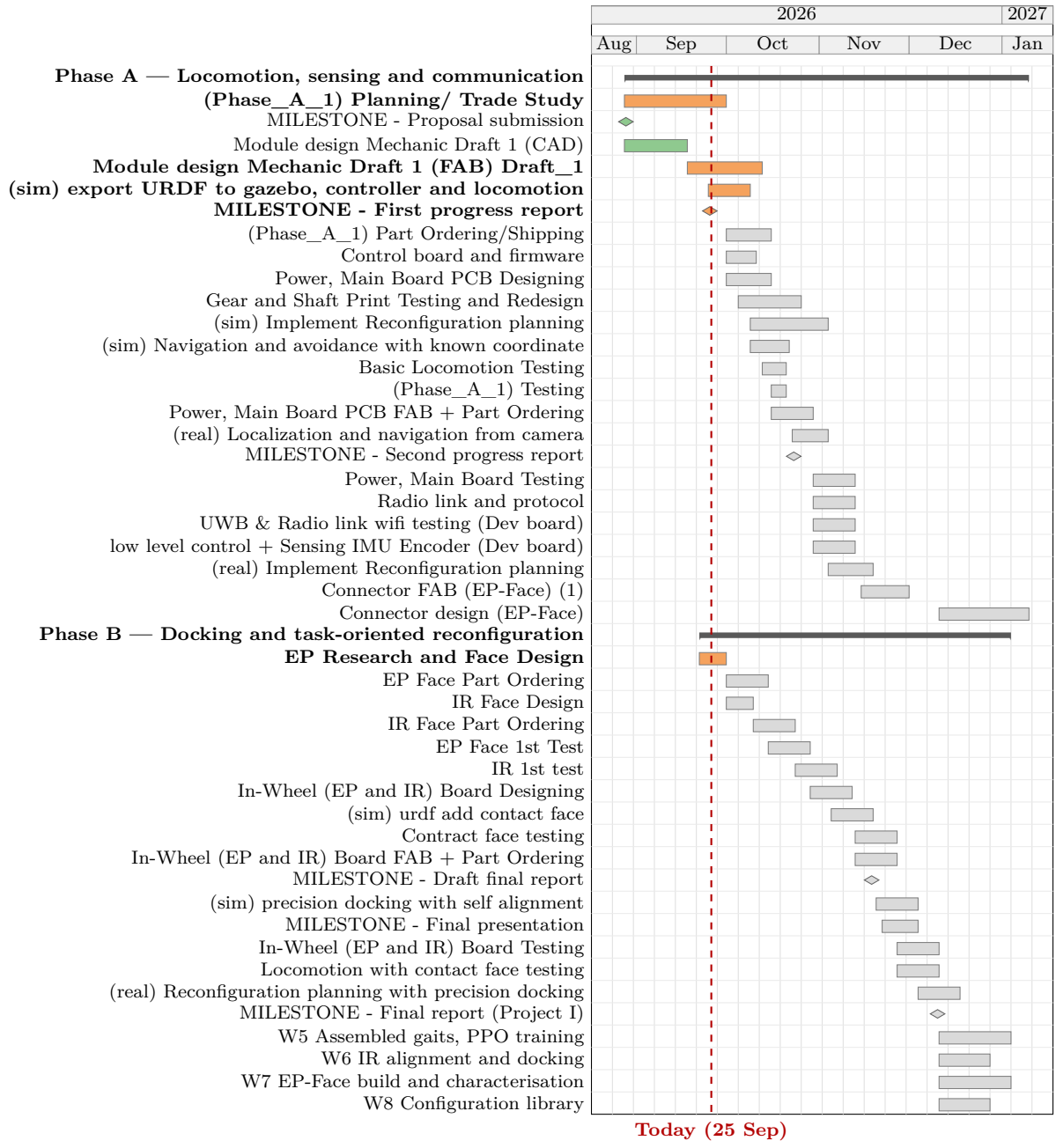


Figure 1: Project I schedule (Phases A–B). Completed In progress Upcoming; dashed line marks the report date.

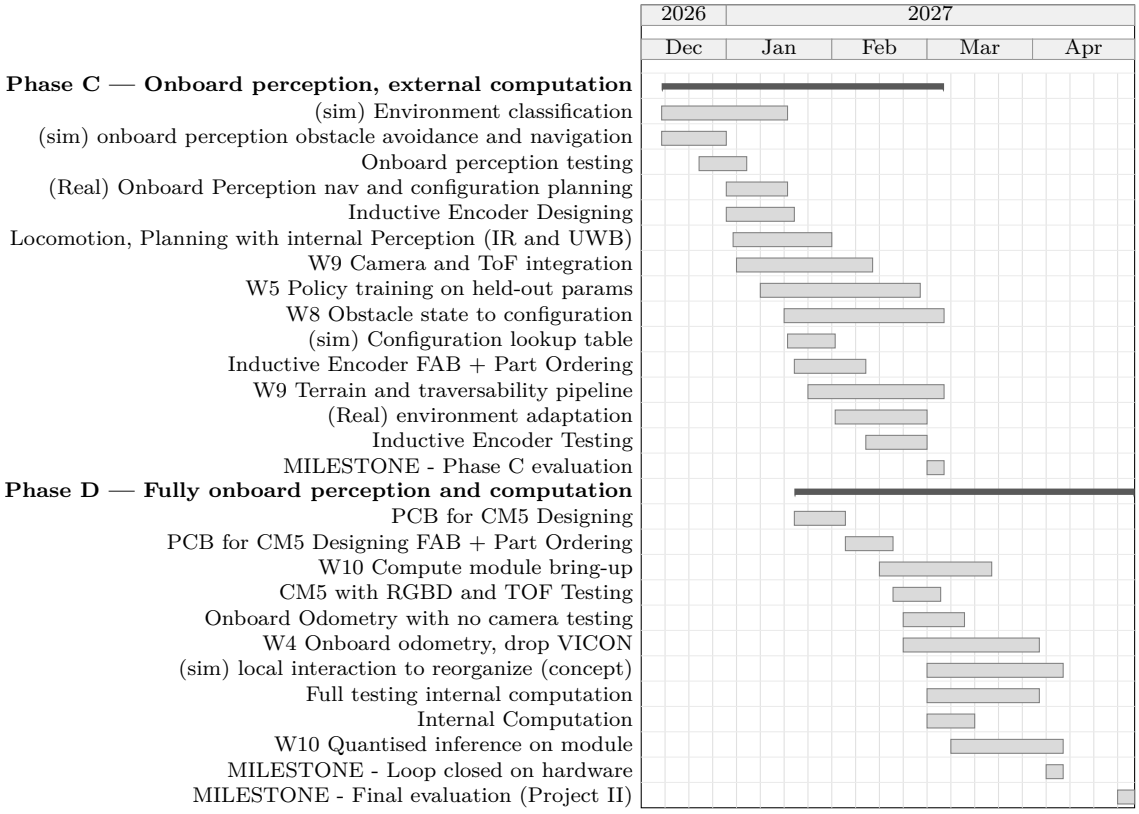


Figure 2: Project II schedule (Phases C–D). All tasks upcoming (▭).

4 Module CAD Design

The design is based on SMORES-EP [1] (originally 80 mm), scaled to a $100 \times 100 \times 100$ mm envelope for additional packaging space.

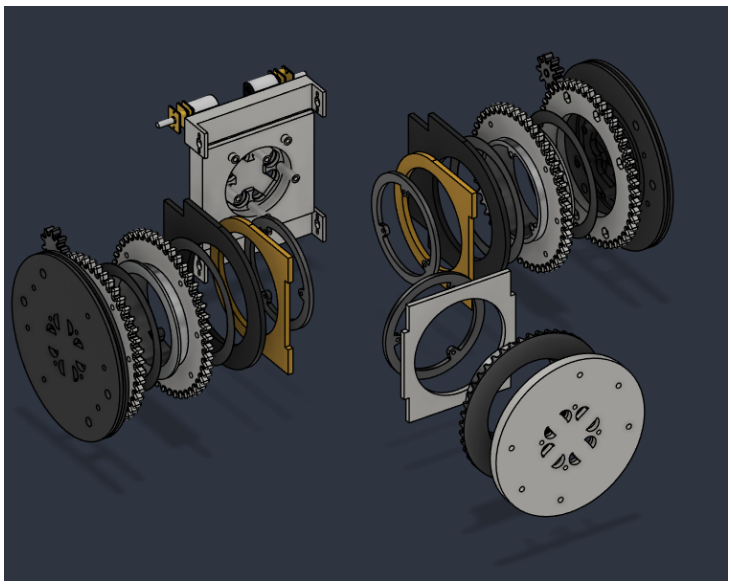
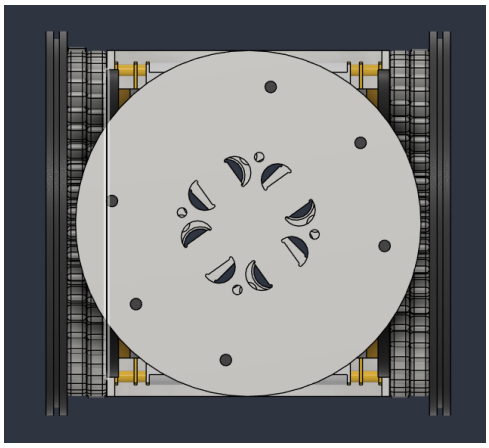
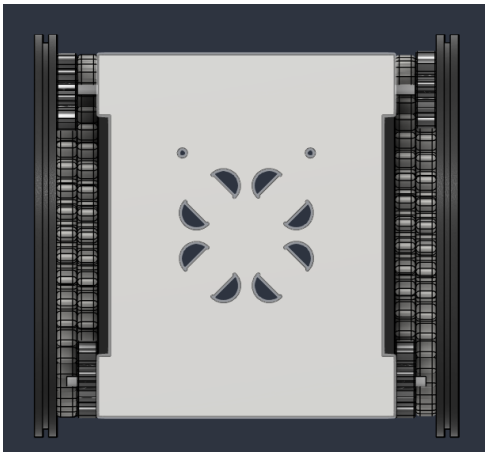


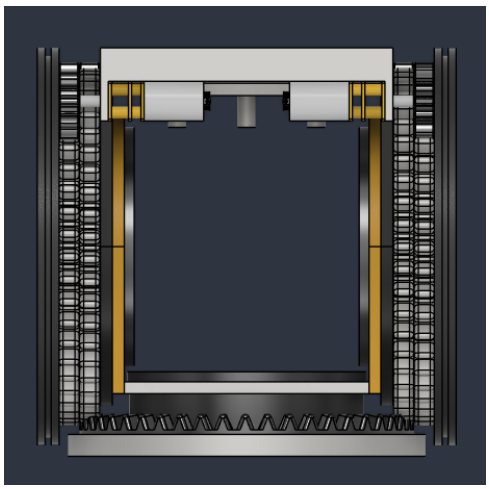
Figure 3: Exploded view of the module.



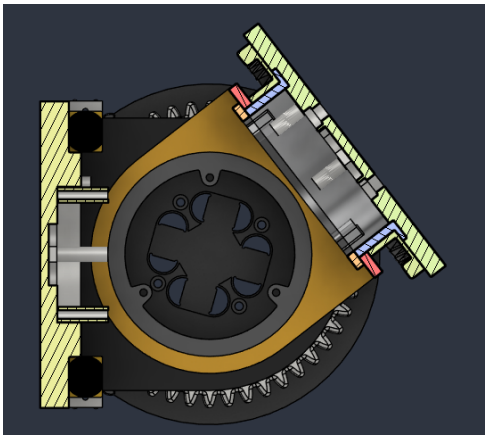
(a) Front view.



(b) Rear view.



(c) Top view.



(d) Tilted sectional view.

Figure 4: Module CAD views.

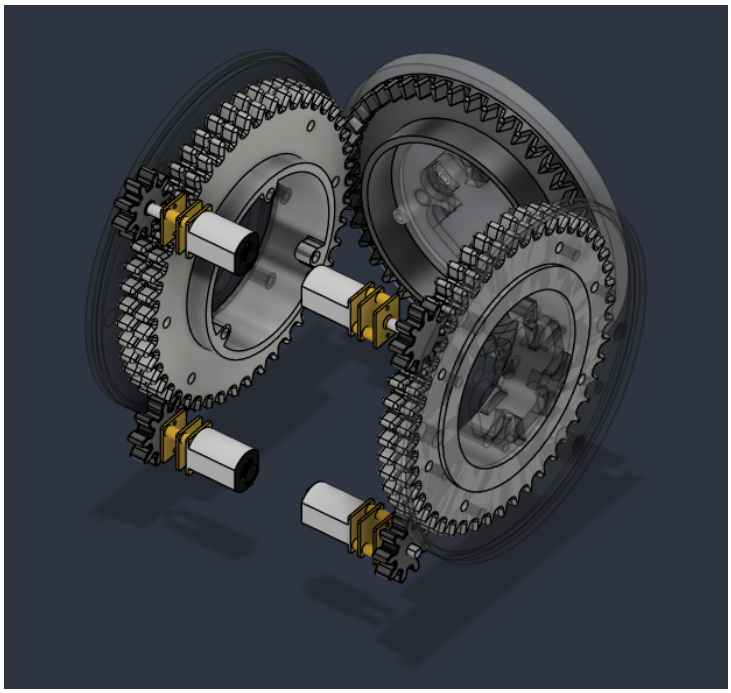


Figure 5: CAD view of the gear train and motor arrangement.

4.1 Gear Train Design

4.1.1 SMORES-EP Reference Baseline

The original SMORES-EP gear train was reverse-engineered from published data (Table 1), yielding module $m = 1.4$, pitch diameter 67.2 mm, and stage ratio $48/9 = 5.33$.

Table 1: Reconstructed SMORES-EP reference parameters.

Parameter	**SMORES-EP**
Module envelope	$80 \times 80 \times 80$ mm
Spur gear	48 teeth
Crown gear	Same part as spur, used at 90°
Pinion	9 teeth
Gearbox	298:1

4.1.2 New Gear Geometry

The 100 mm envelope allows a max spur gear OD of 90 mm. The 5.33:1 ratio was preserved (64/12 teeth) to keep the gearbox selection, SMORES-EP benchmarks, and control firmware valid. The 64/12 pair was chosen over 48/9 because 12 teeth reduces undercutting, improves contact ratio, and provides a larger pinion bore.

Unlike SMORES-EP (which reuses a spur gear as a crown gear), this design uses a **true crown (face) gear** for better load distribution. The crown gear was sized to 84 mm OD (46 teeth, $m = 1.75$) after a 90 mm first attempt interfered with surrounding components.

This creates a **module mismatch** ($m = 1.36$ spur vs. $m = 1.75$ crown) — an accepted compromise since both diameters and tooth counts are fixed by packaging and ratio constraints. The crown dedendum was reduced to 2.15 mm (from standard 2.19 mm) to clear tip interference.

Table 2: Final gear train parameters.

Component	**Teeth**	**Module**	**OD**	**Notes**
Wheel spur gear	64	1.36	90 mm	5 mm face width, bolt holes in web
Pinion (both)	12	1.36	19.04 mm	Through-bore (pan/tilt) or blind (wheel)
Crown gear	46	1.75	84 mm	Dedendum 2.15 mm; true face-gear

Common: 20° pressure angle, 0.2 mm backlash, 0.3 mm root fillet, 0.1 mm bore tolerance. All gears generated via Fusion 360. Material: 3D-printed PETG.

4.1.3 Drivetrain Architecture

Four motors drive four spur gears:

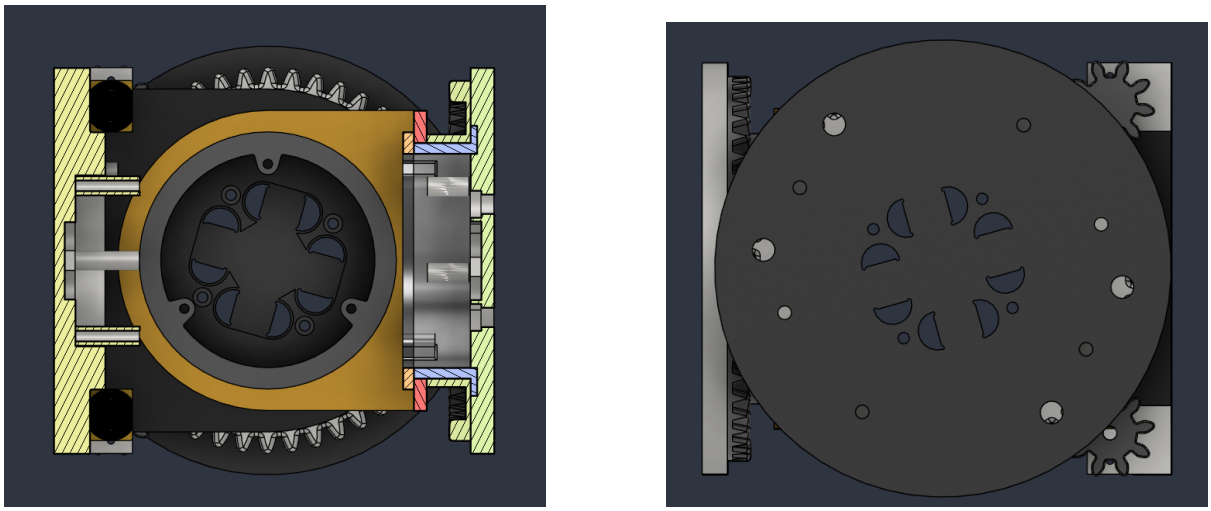
- **Outer two** — each drives a wheel (differential-drive locomotion).
- **Inner two** — both mesh with a single crown gear: same direction → **pan**; opposite directions → **tilt**.

4.1.4 Status

All geometry is CAD-only. **No parts have been printed or tested.** Priority next steps: print first articles, verify spur/crown mesh despite module mismatch, check 12-tooth pinion undercut at print tolerances, and validate backlash and bore fit.

4.2 Outer Wheels

Two independently driven wheels (100 mm diameter) with NBR N70 O-ring tire groove (1.5 mm cross-section), EP-magnet docking space, and PCB mounting posts.



(a) Interior view (wheel and drivetrain). **(b)** Exterior view (wheel and module).

Figure 6: Outer wheel CAD.

4.3 Pan Face

90 mm diameter, no O-ring groove. Includes EP-magnet docking space.

4.4 Tilt Face

Approximately 90 × 71.6 mm rectangular part with EP-magnet docking space and an additional motor mount.

5 Electronics

5.1 Phased Electronics Plan

Electronics are built incrementally across project phases:

- **Phase A-1:** COTS parts — 2S battery, 4-channel motor driver with MCU, ESP32-C3 for Wi-Fi.
- **Phase A:** Custom PCBs — power board (buck, LDO, 4× H-bridge with current sensing) and main board (STM32G474, IMU, UWB, Wi-Fi).
- **Phase B:** In-wheel PCB with EP-Face circuitry and IR contact-alignment, connected via slip ring.
- **Phase C:** RGB-D/ToF camera (Sipeed MaixSense A075V) + wheel encoder added; compute still external.
- **Phase D:** Raspberry Pi CM5 on custom carrier PCB moves compute onboard.

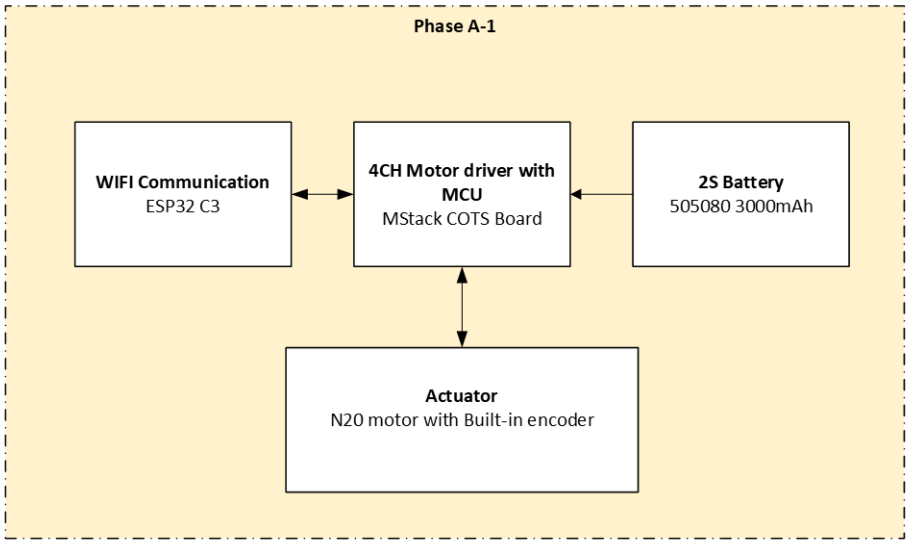
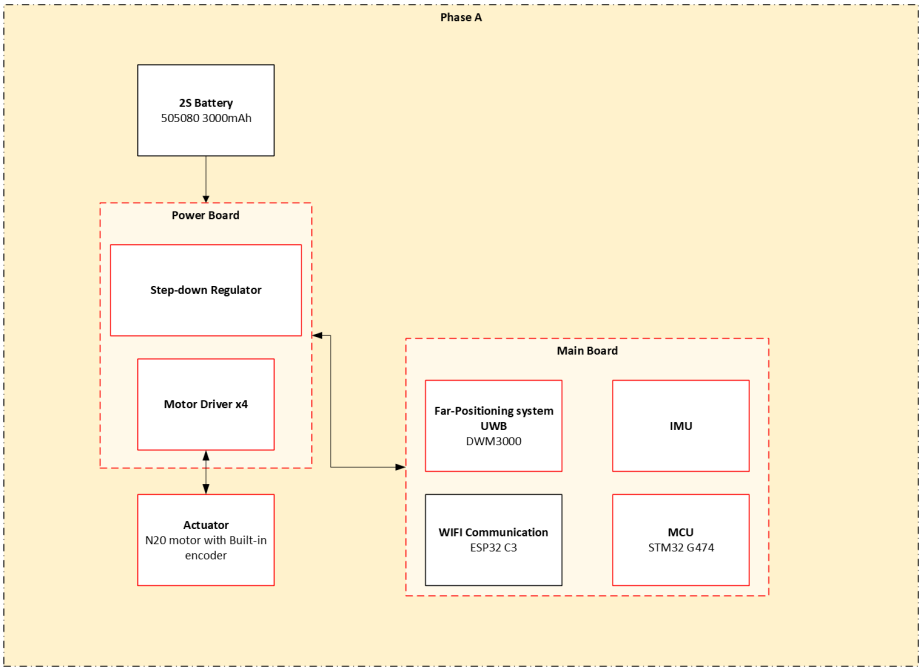
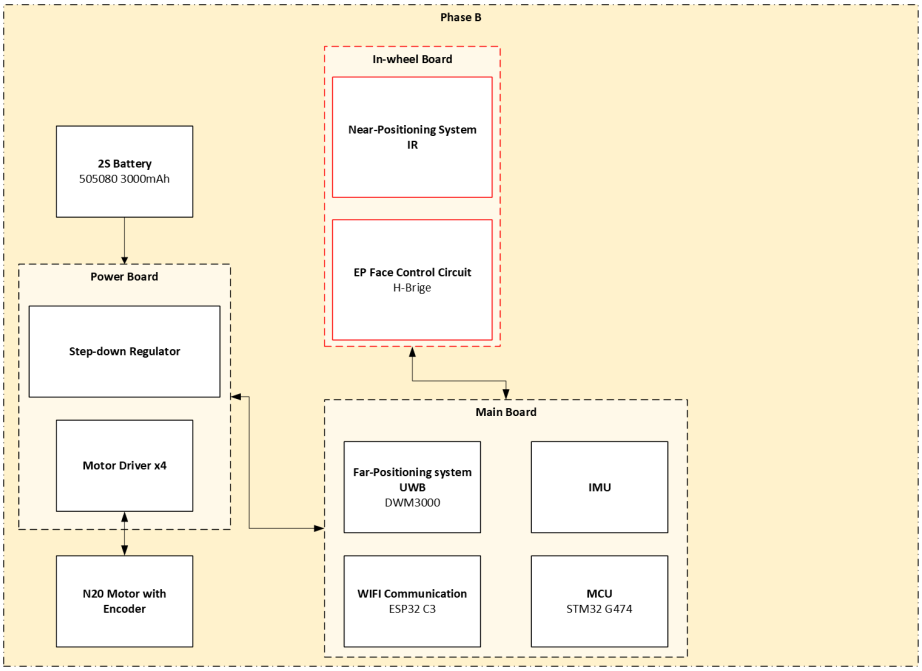


Figure 7: Phase A-1 electronics block diagram.

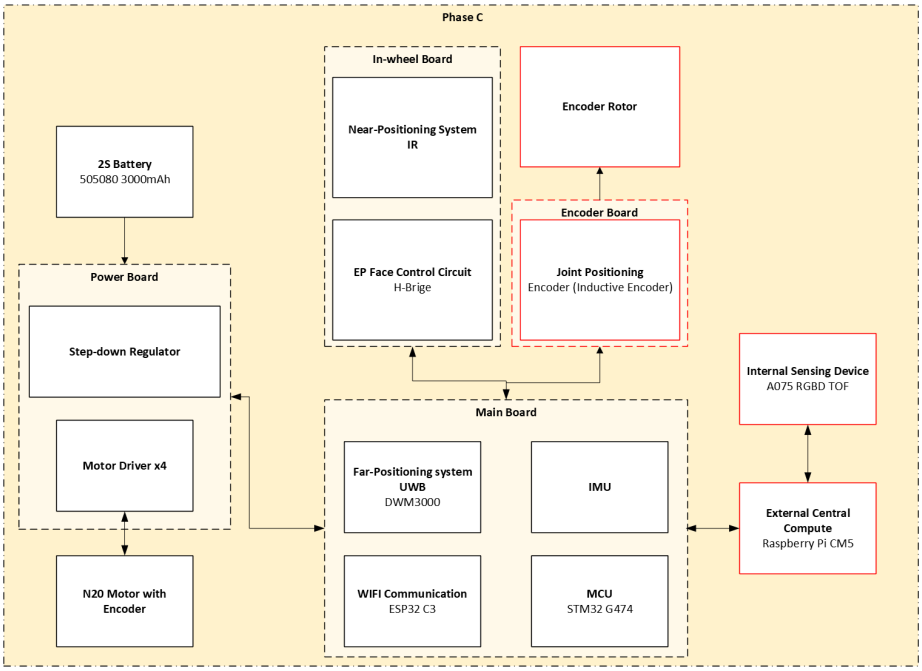


(a) Phase A.

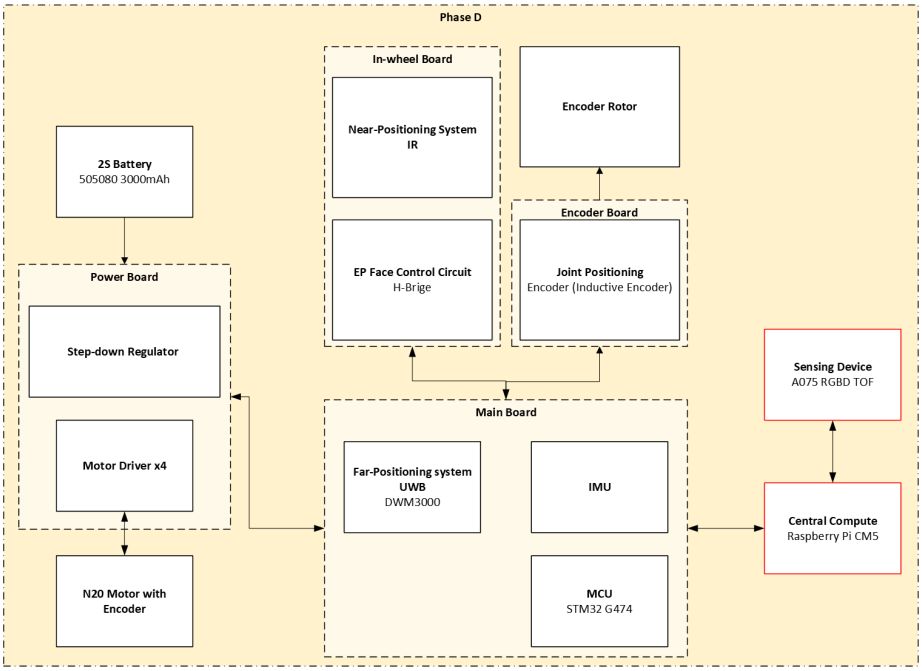


(b) Phase B.

Figure 8: Electronics block diagrams, Phases A–B.



(a) Phase C.



(b) Phase D.

Figure 9: Electronics block diagrams, Phases C–D.

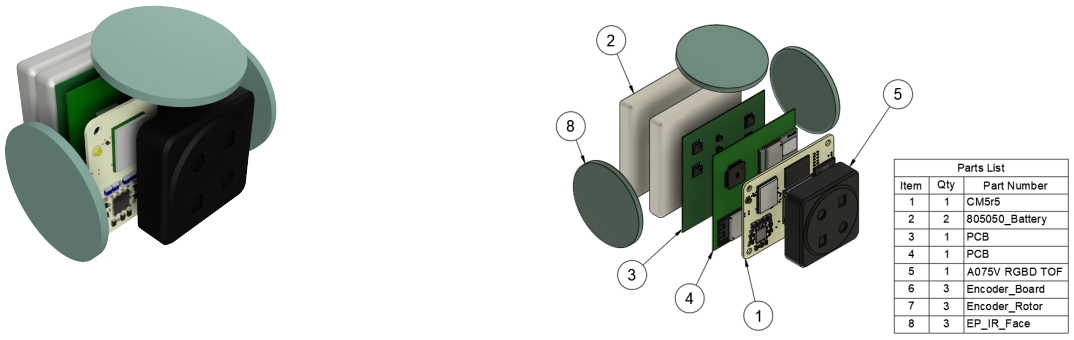
5.2 Component Selection

Table 3: Key electronics components.

Component	Phase	Key Specs	Interface / Supply
4× N20 gearmotor	A-1+	6 V, 50 rpm; 7 PPR encoder	A/B quadrature; H-bridge
STM32G474	A+	Cortex-M4F @ 170 MHz; 12-bit ADC; FDCAN	SPI/I2C/UART/CAN; 3.3 V
ICM-42688-P	A+	±2000 dps / ±16 g; 32 kHz ODR	SPI or I2C; 1.8 V
DW3000 (UWB)	A+	IEEE 802.15.4z; ∼10 cm ranging	SPI; 3.3 V
ESP32-C3	A-1+	Wi-Fi 4 + BLE 5; RISC-V @ 160 MHz	UART/SPI; 3.3 V
EP-Face + IR PCB	B	Custom; EP magnet + IR alignment	Slip ring; 3.3 V
RPi CM5	D	Cortex-A76 @ 2.4 GHz; up to 16 GB RAM	Custom carrier; 5 V
Sipeed MaixSense A075V	C+	ToF 320×240 @ 30 fps; RGB 800×600	USB 2.0; 5 V

5.3 Electronics Packaging

All components fit within the 100 mm cube based on rough CAD placement (Figure 10).



(a) Assembled. **(b)** Exploded.

Figure 10: Electronics packaging.

5.4 Wheel Encoder

Replacing the original continuous-rotation potentiometer (∼5° resolution, contact wear) with a non-contact encoder. Three options considered:

Table 4: Wheel encoder options.

	On-axis magnetic (AS5600)	**Off-axis tape**	**Custom inductive**
Output	Absolute, 12–14 bit	Incremental + index	Absolute
EP interference	Sensitive	Sensitive	Immune
Design effort	Low	Medium	High

The inductive option best handles the EP connector's nearby field but has the highest schedule risk. Decision deferred to Phase C.

5.5 Power System

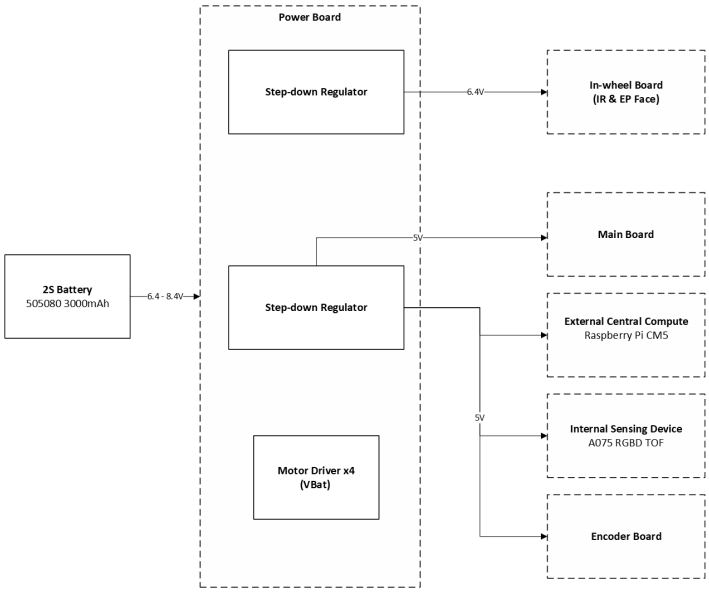


Figure 11: Power distribution block diagram.

Power from a 2S LiPo (3000 mAh, 22.2 Wh) splits three ways:

- **Motors:** 4× N20 via H-bridges, PWM capped at ∼70 % for 6 V rating.
- **EP-Face:** Dedicated buck regulator isolates pulse transients from logic rail.
- **Logic/compute:** Buck to 5 V (CM5, camera); LDO to 3.3 V (MCU, sensors, radios).

Table 5: Power budget summary.

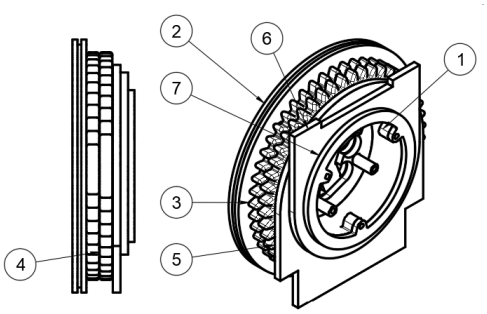
Configuration	**Total from Pack**	**Est. Run Time**
Phase A–B (no compute/camera)	∼3.1 W	**∼5.8 h**
Phase D (full system)	∼11.5 W	**∼1.5 h**

Table 6: Maximum load per rail.

Rail	**Max Current**	**Max Power**
Motor drive (PWM to 6 V)	4.0 A (6 V eq.)	24 W
5 V logic/compute	4.0 A	20 W
3.3 V (from 5 V LDO)	0.45 A	1.5 W
EP-Face + IR	TBD	TBD
Pack total (peak)	∼6.5 A @ 7.4 V	∼47 W

6 Shaft Tolerance Analysis (∅55 mm Bearing)

The plain bearing (OD = 55 mm) acts as the shaft. All PETG-printed components have bores around it.



Parts List		
Item	Qty	Part Name
1	1	Wheel plain bearing v6
2	1	Wheel (left and right) v19
3	2	Spur gear v17
4	2	Shim Ring 55 68 0.2 v2
5	1	Lower-tilt-joint v12
6	1	Upper-tilt-joint v8
7	1	Bushing block v3

Figure 12: Shaft and bearing component arrangement.

Table 7: Joint classification and ISO fits for ∅55 mm bearing.

Joint	Component	Motion	Fit	ISO Class
J1	Wheel + 1st Spur Gear	Dynamic	Clearance	F7/h6
J2	2nd Spur Gear (Pan/Tilt)	Dynamic	Clearance	F7/h6
J3	Upper Tilt Joint	Dynamic	Clearance	F7/h6
J4	Lower Tilt Joint	**Static**	**Interference**	P7/h6

ISO 286 limits (bearing h6: 55.000–54.981 mm): Dynamic F7 bore 55.030–55.060 mm (clearance +0.030 to +0.079 mm); Static P7 bore 54.938–54.968 mm (interference +0.013 to +0.062 mm).

FDM print compensation: Printed holes come out undersized, so the ISO limits cannot be used directly as CAD dimensions. The analysis above was used as the starting point for a printed-fit test. PETG was chosen over nylon because it prints easily and the mechanical design is not a focus of this project.

Test result: A CAD bore of ∅55.00 + 0.15 = ∅55.15 mm was tested on the printed PETG parts and works. This single offset is adopted for **all** components on the shaft (wheel, spur gears, upper and lower tilt joints), replacing the separate dynamic and static bore sizes originally proposed (Table 8).

Table 8: Final CAD bore size (tested).

Part	Joint	CAD Bore (mm)
Wheel / Spur Gears / Upper Tilt	J1–J3	∅55.15
Lower Tilt Joint	J4	∅55.15

Bearing OD = ∅55.00 mm; offset +0.15 mm on all bores.

7 Teleoperation Architecture

Figure 13 shows the teleoperation stack. Controllers, topics and the robot model are shared between simulation and hardware; a launch argument selects the `ros2_control` hardware plugin.

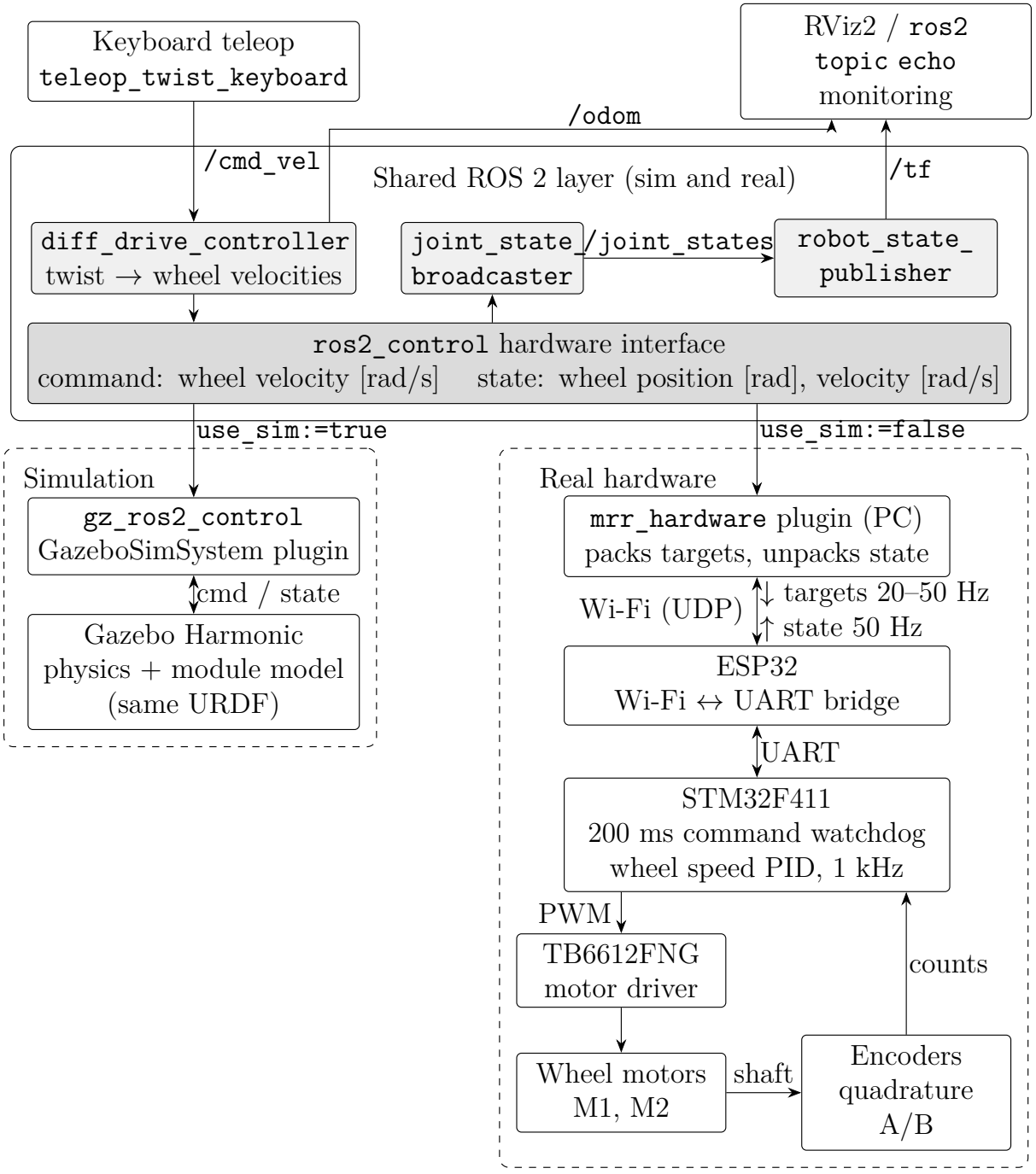


Figure 13: Teleoperation architecture. The controllers, topics and robot model are shared; a launch argument selects the hardware plugin under the `ros2_control` interface, either Gazebo through `gz_ros2_control` or one physical module over Wi-Fi.

8 Software Testing Plan (Phase A)

8.1 Objectives

Verify the integrated software pipeline: AprilTag detection and robot ID via external camera → real-time pose estimation → PID motor control → path planning and execution for MSRR configurations.

8.2 Setup

MSRR robot in a controlled indoor environment with an external camera, AprilTags for ID and pose reference, and a computing system running perception, localisation, planning, and control.

8.3 Goal Configurations

Snake, Car, DoubleDriver, L-shape.

8.4 Evaluation Metrics

- **Detection rate** — % frames with correct AprilTag ID.
- **Position / orientation error** — estimated vs. reference pose.
- **Tracking error** — desired vs. actual position and heading during path following.
- **Control response** — PID overshoot and steady-state error.
- **Processing rate** — perception + control update frequency.

8.5 Acceptance Criteria

Reliable tag detection, acceptable localisation and tracking error for real-time control, stable PID behaviour without excessive oscillation, and continuous operation without critical failures. Specific numerical thresholds to be defined based on system requirements.

9 Preparation for Phase B: EP Magnet

The docking system uses an Electro-Permanent Magnet (EPM) connector based on SMORES-EP [1]. Each EP unit pairs an AlNiCo 5 magnet (low coercivity, switchable) with an NdFeB magnet (high coercivity, fixed) between steel pole pieces, wound with a copper coil.

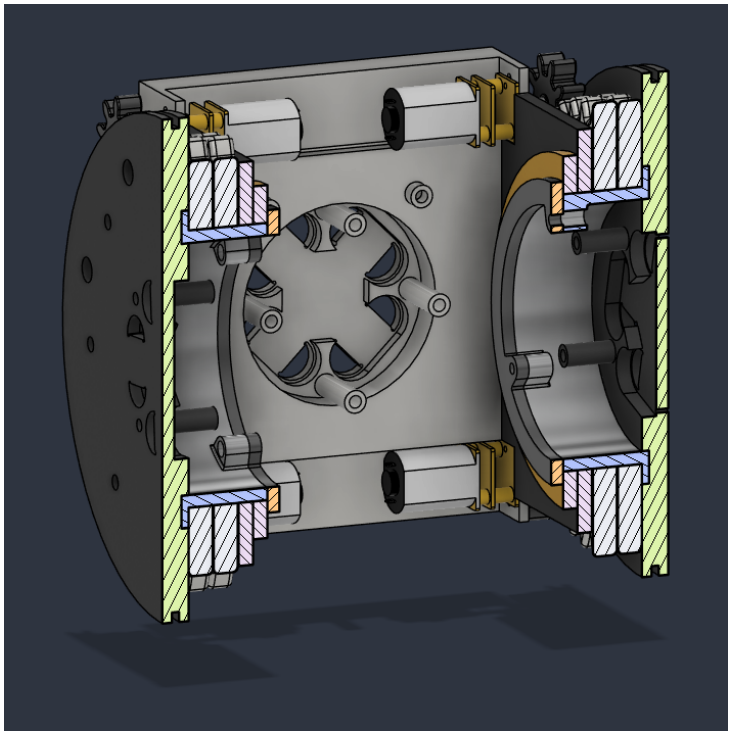


Figure 14: Interior view of the rectangular magnet housing.

Operating principle: A current pulse through the coil reverses AlNiCo 5 magnetisation while NdFeB stays fixed. Fields aligned → **ON** (attract); fields opposed → **OFF** (release). No power needed to hold either state.

Materials: AlNiCo 5 rod, NdFeB magnet, mild steel 1018/S20C pole pieces, 30 AWG enamelled copper wire (200 turns). Original dimensions (4.76 mm dia. × 9.53 mm, four magnets in 42 mm ring) being adapted for the 100 mm face. Final dimensions and driving circuit to be set in Phase B testing.

Additional References

[1] Tosun, T., Davey, J., Liu, C., & Yim, M. (2016). Design and characterization of the EP-Face connector. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 45–51).